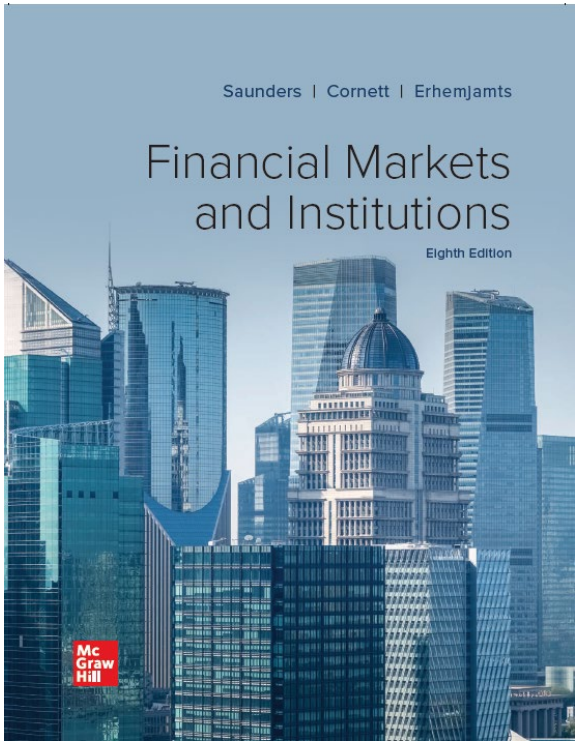
# Chapter Four

# The Federal Reserve System, Monetary Policy, and Interest Rates

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# Major Duties and Responsibilities of the Federal Reserve System: Chapter Overview

- **Central banks** determine, implement, and control the monetary policy in their home countries
- The Federal Reserve (the Fed) is the central bank of the United States
  - Founded by Congress under Federal Reserve Act of 1913
  - Independent central bank
  - Duties incorporate four major functions:
    1. Conducting monetary policy
    2. Supervising and regulating depository institutions
    3. Maintaining the stability of the financial system
    4. Providing payment and other financial services to the U.S. government, the public, financial institutions, and foreign official institutions

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# Structure of the Federal Reserve System (FRS)

- 12 Federal Reserve Banks in cities throughout the U.S.
  - Each acts as a depository institution for the banks in its district
  - Operate under general supervision of the Board of Governors
  - Each has its own 9-member board of directors
  - Nationally chartered banks are required to become members of the FRS
- 7-member Board of Governors in Washington, D.C.
  - Each member appointed by the president of the U.S. and must be confirmed by the Senate
  - Members serve a nonrenewable, 14-year term
  - Primary responsibilities are the formulation and conduct of monetary policy and the supervision and regulation of banks

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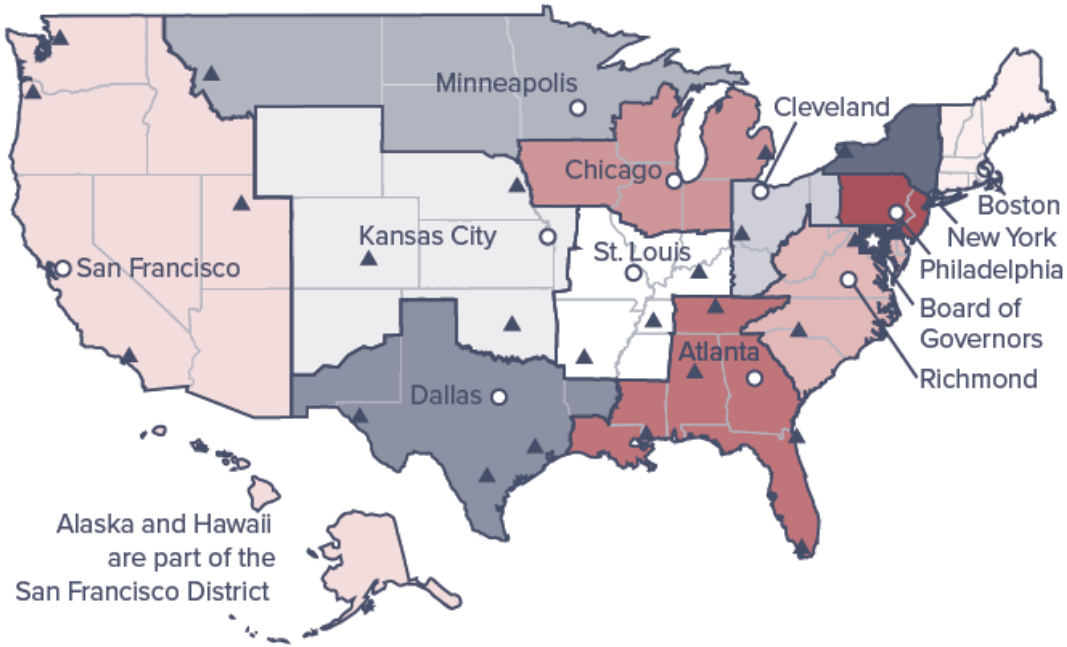
# Federal Reserve Districts

**Figure 4–1** Federal Reserve Districts




- Board of Governors of the Federal Reserve System, Washington, DC
- Federal Reserve Bank city
- Federal Reserve Branch city, by District

| | |
|---|---|
| Buffalo | Little Rock, Louisville, Memphis |
| Cincinnati, Pittsburgh | Helena |
| Baltimore, Charlotte | Denver, Oklahoma City, Omaha |
| Birmingham, Jacksonville, Miami, Nashville, New Orleans | El Paso, Houston, San Antonio |
| Detroit | Los Angeles, Portland, Salt Lake City, Seattle |

# Federal Open Market Committee (FOMC)

- **FOMC** is the major monetary policy-making body of the Federal Reserve System
  - Required to meet at least four times each year in Washington, D.C., but typically meet more often
  - Main responsibilities are to formulate policies to promote full employment, economic growth, price stability, an da sustainable pattern of international trade
  - Set guidelines regarding **open market operations**, the purchase and sale of U.S. government and federal agency securities, as the main policy tool to achieve monetary targets
  - Beige Book summarizes information on current economic conditions by Federal Reserve district

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# Functions Performed by Federal Reserve Banks (FRBs)

1. Assistance in the conduct of monetary policy
   - Set and change the **discount rate**
   - Loans transacted through each FRBs **discount window**
2. Supervision and regulation
   - Each FRB has supervisory and regulatory authority over the activities of state-chartered member banks and bank holding companies located in their districts
3. Consumer protection and community affairs
   - Possess authority to implement federal laws intended to protect consumers in credit and other financial transactions
4. Government services
   - Serves as the commercial bank for the U.S. Treasury

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# Functions Performed by Federal Reserve Banks (FRBs) (Continued)

5. New currency issue
   - Responsible for the collection and replacement of currency (paper and coin) from circulation
6. Check clearing
   - Operates a central check clearing system for U.S. banks, routing interbank checks to DIs on which they are written and transferring the appropriate funds from one bank to another
7. Wire transfer services
   - FRBs and member banks are linked electronically through the Federal Reserve Communications System
8. Research services
   - Each FRB uses professional economics to conduct research

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# Balance Sheet of the Federal Reserve

- Liabilities
  - Major liabilities on the Fed's balance sheet are currency in circulation and **reserves**, the sum of which is referred to as the Fed's **monetary base** or **money base**
  - Total reserves can be classified into two categories:
    - **Required reserves** are those the Fed requires banks to hold by law
    - **Excess reserves** are additional over and above required reserves
- Assets
  - Major assets are Treasury and government agency (i.e., Fannie Mae, Freddie Mac) securities, Treasury currency, and gold and foreign exchange
  - Interbank loans are a small portion of total assets, but they plan an important role in implementing monetary policy

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# Balance Sheet of the Federal Reserve (Continued)

| Assets | 2007 | 2008 | 2010 | 2013 | 2016 | 2018 | Percentage of Total, 2018 |
|---|---|---|---|---|---|---|---|
| U.S. official reserve assets | $ 34.2 | $ 35.7 | $ 37.0 | $ 34.5 | $ 30.5 | $ 31.80 | 0.8% |
| SDR certificates | 2.2 | 2.2 | 5.2 | 5.2 | 5.2 | 5.20 | 0.1 |
| Treasury currency | 38.7 | 38.7 | 43.5 | 45.0 | 47.6 | 49.80 | 1.2 |
| Federal Reserve float | −0.0 | −1.5 | −1.4 | −0.6 | −0.0 | −0.80 | 0.0 |
| Interbank loans | 48.6 | 559.7 | 0.2 | 0.0 | 0.1 | 0.10 | 0.0 |
| Security repurchase agreements | 46.5 | 80.0 | 0.0 | 0.0 | 0.0 | 0.00 | 0.0 |
| U.S. Treasury securities | 740.6 | 475.9 | 1,021.5 | 1,796.0 | 2,461.6 | 2,338.00 | 55.2 |
| U.S. government agency securities | 0.0 | 19.7 | 1,139.6 | 1,143.4 | 1,780.4 | 1,644.60 | 38.8 |
| Miscellaneous assets | 40.5 | 1,060.2 | 207.6 | 220.3 | 216.8 | 167.40 | 4.0 |
| Total assets | $951.3 | $2,270.6 | $2,453.2 | $3,243.8 | $4,542.2 | $4,236.00 | 100.0% |
| **Liabilities and Equity** | | | | | | | |
| Depository institution reserves | $ 20.8 | $ 860.0 | $ 968.1 | $1,790.4 | $1,977.2 | $1,556.00 | 36.7% |
| Vault cash of commercial banks | 55.0 | 57.7 | 52.7 | 59.7 | 74.2 | 82.00 | 1.9 |
| Deposits due to federal government | 16.4 | 365.7 | 340.9 | 79.4 | 333.7 | 402.40 | 9.5 |
| Deposits due to government agencies | 1.7 | 21.1 | 13.5 | 20.2 | 31.1 | 38.50 | 0.9 |
| Currency outside banks | 773.9 | 832.2 | 930.0 | 1,117.3 | 1,350.8 | 1,637.20 | 38.6 |
| Security repurchase agreements | 44.0 | 88.4 | 59.7 | 105.5 | 712.4 | 304.00 | 7.2 |
| Miscellaneous liabilities | 2.5 | 3.4 | 35.3 | 16.2 | 23.3 | 81.70 | 1.9 |
| Federal Reserve Bank stock | 18.5 | 21.1 | 26.5 | 27.6 | 29.5 | 32.30 | 0.8 |
| Equity | 18.5 | 21.0 | 26.5 | 27.5 | 10.0 | 101.90 | 2.4 |
| Total liabilities and equity | $951.3 | $2,270.6 | $2,453.2 | $3,243.8 | $4,542.2 | $4,236.00 | 100.0% |

**Source:** Federal Reserve Board, "Financial Accounts of the United States - Z.1," *Monetary Authority,* December 2019, p. L.109, https://www.federalreserve.gov

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# Monetary Policy Tools

- Federal Reserve uses the following to implement monetary policy:
  - Open market operations
  - Discount rate
  - Reserve requirements
- Major link by which monetary policy impacts the macroeconomy occurs through the Federal Reserve influencing the market for bank reserves
- Federal Reserve's monetary policy seeks to influence either the demand for, or supply of, excess reserves at depository institutions and in turn the money supply and the level of interest rates

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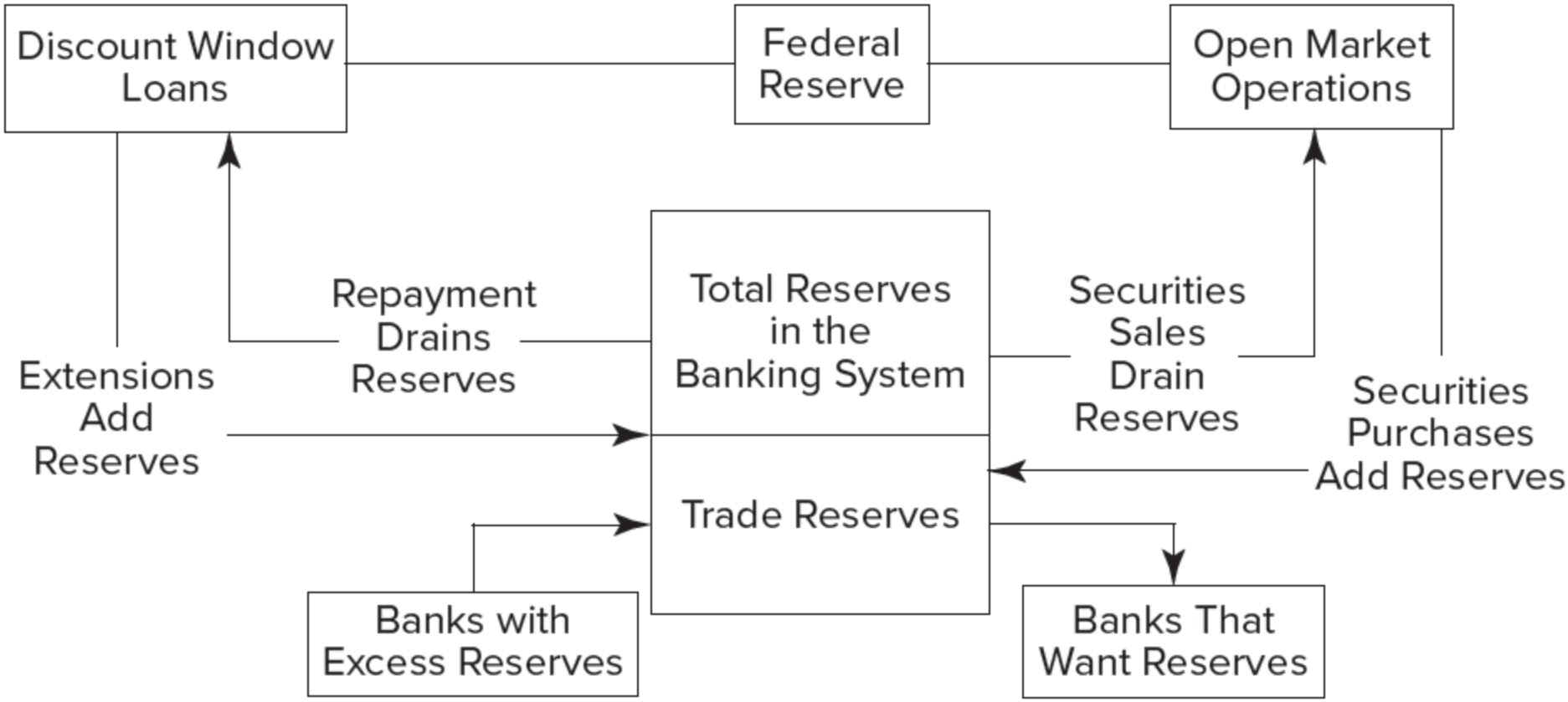
# Federal Reserve Monetary Policy Activities

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# Monetary Policy Tools (Continued)

- Depository institutions trade excess reserves held at their local Federal Reserve Banks among themselves
  - Rate of interest (or price) on these interbank transactions is a benchmark interest rate, called the **fed funds rate**
- Financial Services Regulatory Relief Act of 2006 authorized the Federal Reserve to pay interest on reserve balances
  - Interest on excess reserves (IOER) and required reserves (IORR)
- Federal Reserve can take one of two basic approaches to affect the market for banks' excess reserves
  1. Target the quantity of reserves in the market
  2. Target the interest rate on those reserves (the fed funds rate)

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# Monetary Policy Tools: Open Market Operations

- Open market operations
  - When a targeted monetary aggregate or interest rate level is determined by the FOMC, it is forwarded to the **Federal Reserve Board Trading Desk** at the Federal Reserve Bank of New York (FRBNY) through a statement called the **policy directive**
  - Manager of Trading Desk uses policy directive to instruct traders on daily amount of open market purchases or sales to transact
- Open market operations are the primary determinant of changes in bank excess reserves in the banking system
  - Directly impact the size of the money supply and/or the level of interest rates (e.g., the fed funds rate)
- Open market operations are primarily conducted using Treasury securities, but others can be used as well

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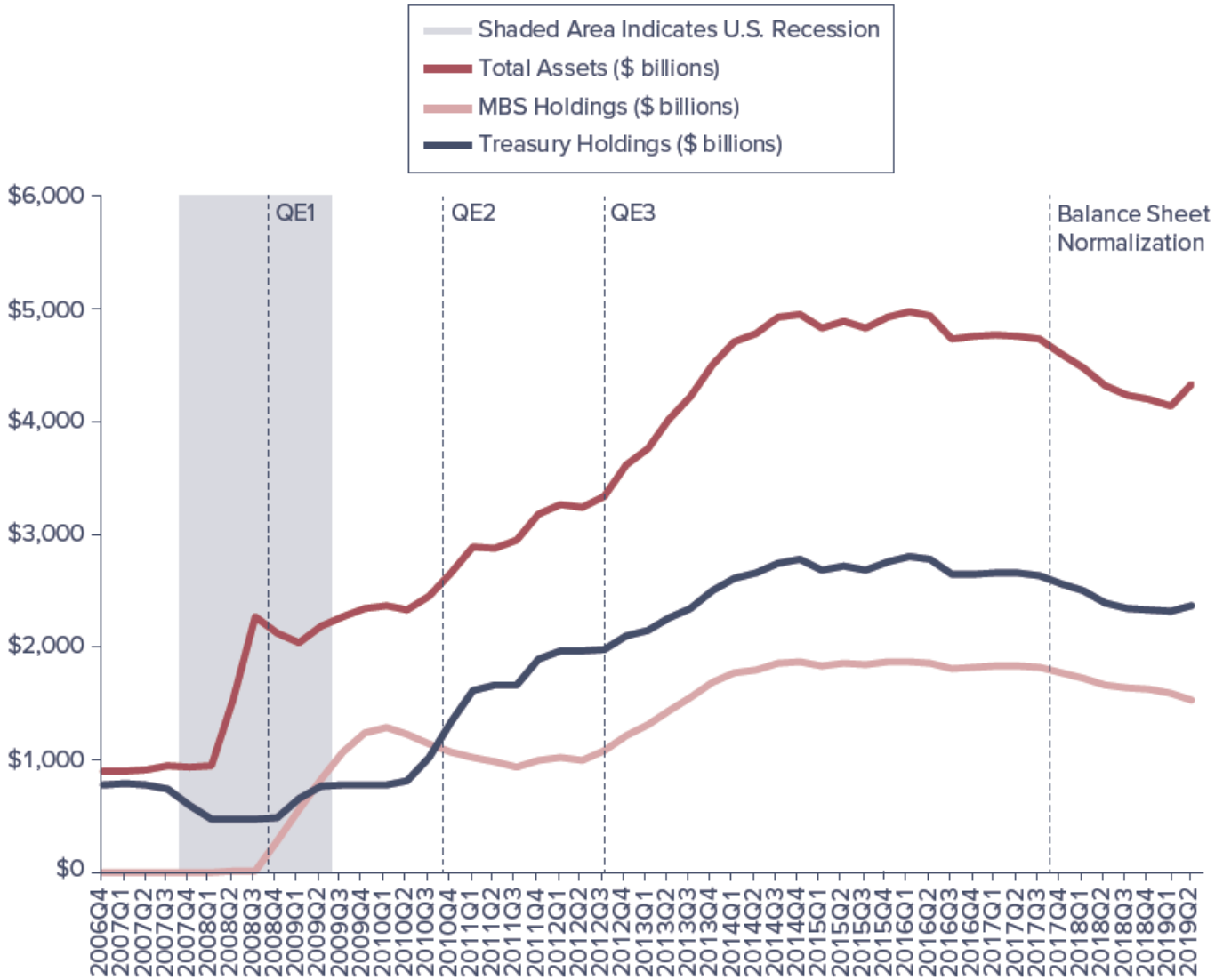
# Federal Reserve's Balance Sheet and Large-Scale Asset Purchases



# Monetary Policy Tools: Discount Rate

- Recall, the discount rate is the rate of interest Federal Reserve Banks charge on loans to FIs in their district
  - Raising the discount rate signals a desire to see a tightening of monetary conditions and higher interest rates in general
  - Lowering the discount rate signals a desire to see more expansionary monetary conditions and lower interest rates in general
- Federal Reserve has rarely used the discount rate as a monetary policy tool for the following reasons:
  1. It is difficult for the Fed to predict changes in bank discount window borrowing when the discount rate changes
  2. Because of its “signaling” importance, a discount rate change often has great effects on the financial markets

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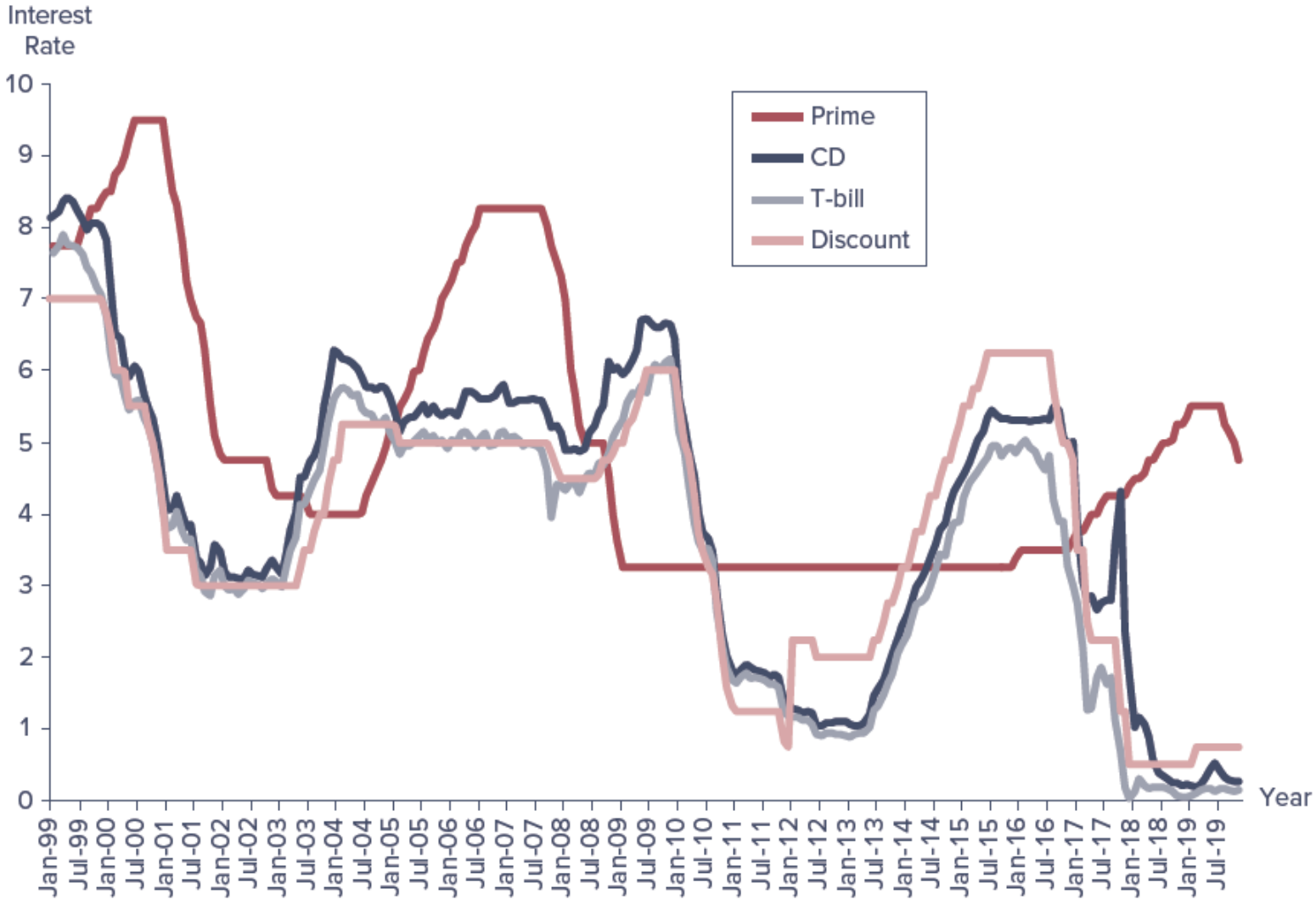
# Various U.S. Interest Rates



# Monetary Policy Tools: Discount Rate (Continued)

- Historically, discount window lending was limited to depository institutions (DIs) with severe liquidity needs
  - Fed made changes to discount window lending that increased the cost of borrowing but eased the terms in January 2003
  - Three lending programs are now offered through the Fed's discount window:
    - *Primary credit* is available to generally sound DIs on a very short-term basis, typically overnight, at a rate above the Federal Open Market Committee's target rate for federal funds
    - *Secondary credit* is available to meet backup liquidity needs when its use is consistent with a timely return to a reliance on market sources of funding or the orderly resolution of a troubled institution
    - *Seasonal credit* is available to DIs that can demonstrate a clear pattern of recurring intrayearly swings in funding needs

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# Reserve Requirements (Reserve Ratios)

- Reserve requirements determine the minimum amount of reserve assets that DIs must maintain by law to back transaction deposit accounts held as liabilities on their balance sheets
  - Requirement is usually set as a ratio of transaction accounts
  - Very rarely used by the Federal Reserve as a monetary policy tool
- A(n) decrease (increase) in the reserve requirement ratio means that DIs may hold fewer (must hold more) reserves against their transaction accounts, allowing them to lend out a greater (smaller) percentage of their deposits and increasing (decreasing) credit availability in the economy

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# Reserve Requirements (Reserve Ratios) (Continued)

- Decrease in the reserve requirement results in a multiplier increase in the supply of bank deposits and thus the money supply
  - Multiplier effect:

$$\text{Change in bank deposits} = (1/\text{New reserve requirement}) \times \text{Increase in reserves created by reserve requirement change}$$

- Increase in the reserve requirement results in a multiple contraction in deposits and a decrease in the money supply
  - Multiplier effect:

$$\text{Change in bank deposits} = (1/\text{New reserve requirement}) \times \text{Decrease in reserves created by reserve requirement change}$$

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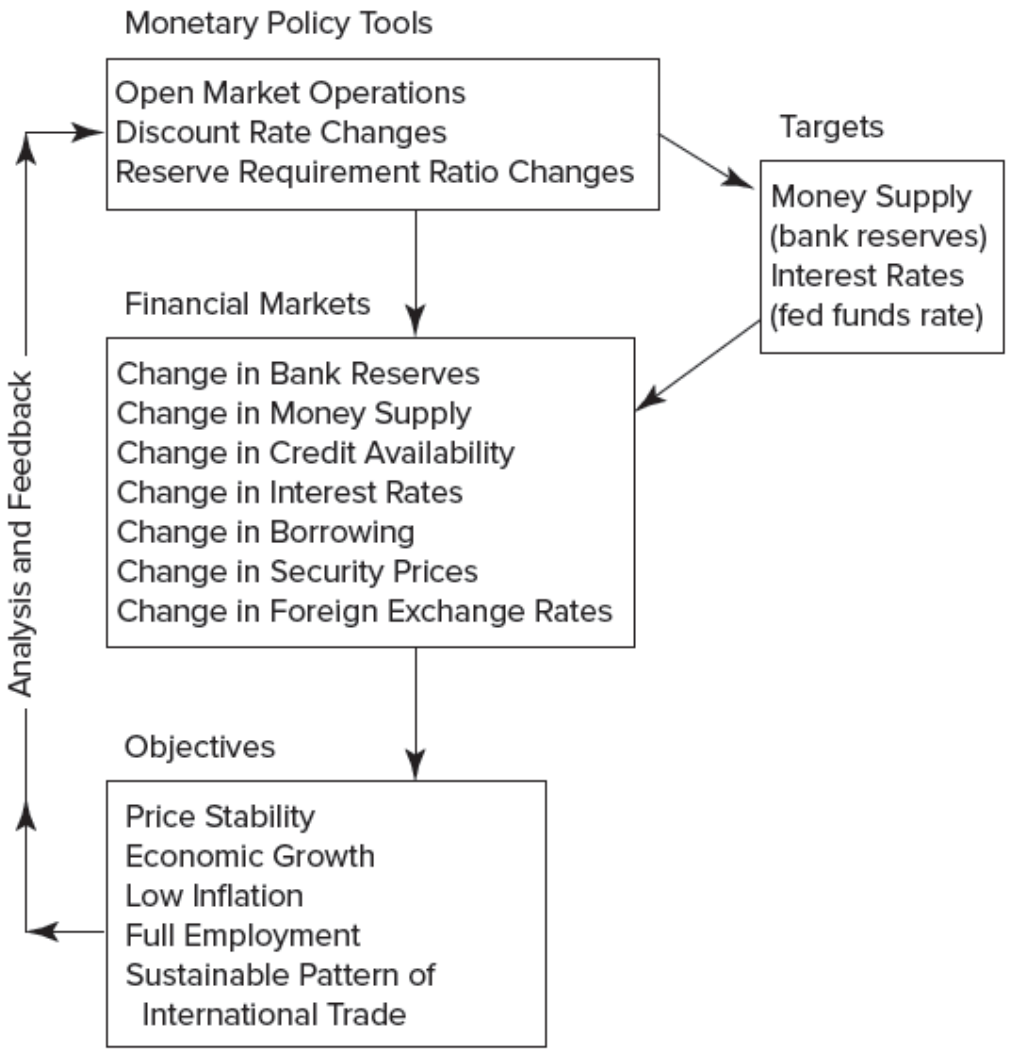
# The Process of Monetary Policy Implementation

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# Effects of Monetary Tools on Various Economic Variables

**Expansionary Activities**

- Open market purchases of securities
  - All else constant, reserve accounts of banks increase
- Discount rate decreases
  - All else constant, interest rates in the economy decrease
- Reserve requirement ratio decreases
  - All else constant, bank reserves increase

**Contractionary Activities**

- Open market sales of securities
  - All else constant, reserve accounts of banks decrease
- Discount rate increases
  - All else constant, interest rates in the open market increase
- Reserve requirement ratio increases
  - All else constant, bank reserves decrease

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# The Impact of Monetary Policy on Various Economic Variables

| | Expansionary Activities (open market purchases of securities, discount rate decreases, reserve requirement ratio decreases) | Contractionary Activities (open market sales of securities, discount rate increases, reserve requirement ratio increases) |
|---|---|---|
| Impact on: | | |
| Reserves | ↑ | ↓ |
| Money supply | ↑ | ↓ |
| Credit availability | ↑ | ↓ |
| Interest rates | ↓ | ↑ |
| Security prices | ↑ | ↓ |

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# Money Supply versus Interest Rate Targeting

- Federal Reserve can successfully target *only one of these two variables* (money supply or interest rates) at any one moment
  - If the money supply is the target variable used to implement monetary policy, interest rates must be allowed to fluctuate relatively freely
  - If an interest rate (e.g., fed funds rate) is the target, bank reserves and the money supply must be allowed to fluctuate relatively freely

| Target | Years |
|---|---|
| Fed funds rate targeted using bank reserves to achieve target | 1970–October 1979 |
| Nonborrowed reserves targeted | October 1979–October 1982 |
| Borrowed reserves targeted | October 1982–July 1993 |
| Fed funds rate targeted (rate announced) | July 1993–present |

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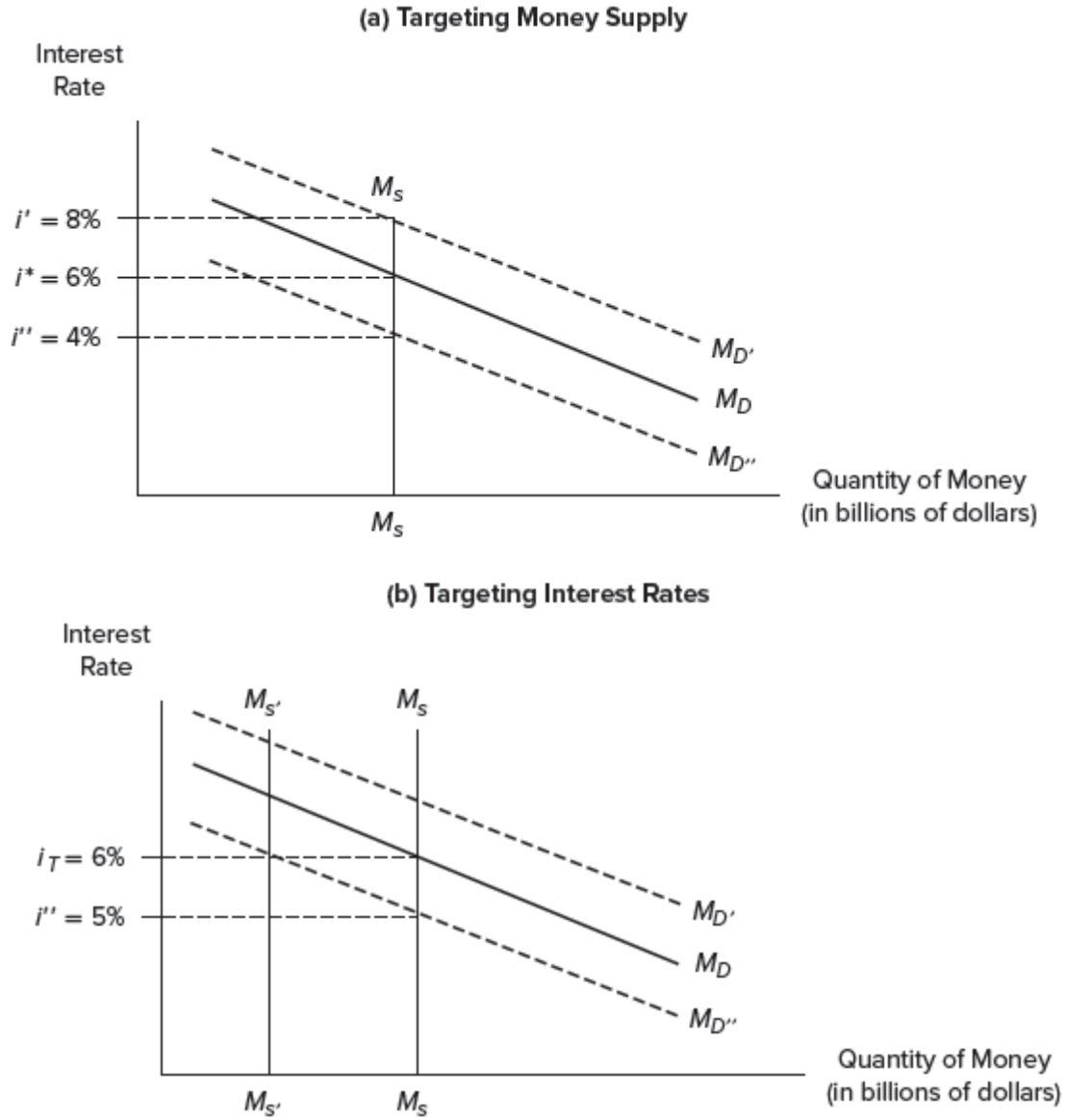
# Federal Reserve Monetary Policy Targets

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# Money Supply versus Interest Rate Targeting (Continued)

- Fed announced that it would use interest rates—the federal funds rate—as the main target variable to conduct monetary policy in 1993
  - Guiding principle used by the Fed to set short-term interest rates is the Taylor rule, which states short-term interest rates should be determined by three conditions:
    1. Where actual inflation is relative to the Fed's targeted level
    2. The extent to which the economy is above or below its full employment level
    3. What short-term interest rates should be to achieve full employment
  - Under the current regime, the Fed simply announces whether the federal funds rate target has been increased, decreased, or left unchanged after every monthly FOMC meeting

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